Qualimap Analysis Results

BAM QC analysis

Generated by Qualimap v.2.3

2026/02/25 15:34:35

1. Input data & parameters

1.1. QualiMap command line

```
qualimap bamqc -bam
userspace/66ebdff874f1165c2d9d9eae8365cb5ac21cdeda/workflow/bsmap/
F6703_372760_human.bam -nw 400 -hm 3
```

1.2. Alignment

Command line:	/public3/home/scg9946/.local/share/mamba/envs/xdxtools-core/bin/STAR-avx2 --runThreadN 40 --genomeDir ./inst/rnaseq/homo_sapiens/ --readFilesIn userspace/66ebdff874f1165c2d9d9eae8365cb5ac21cdeda/workflow/trim/F6703_372760_val_1.fq.gz userspace/66ebdff874f1165c2d9d9eae8365cb5ac21cdeda/workflow/trim/F6703_372760_val_2.fq.gz --readFilesCommand zcat --outFileNamePrefix userspace/66ebdff874f1165c2d9d9eae8365cb5ac21cdeda/workflow/bsmap/human/F6703_372760 --outSAMtype BAM SortedByCoordinate --outSAMattributes NH HI AS nM NM MD --outSAMunmapped None --quantMode GeneCounts --twopassMode Basic
Draw chromosome limits:	no
Analyze overlapping paired-end reads:	no

Program:	STAR (2.7.11b)
Analysis date:	Wed Feb 25 15:34:34 CST 2026
Size of a homopolymer:	3
Skip duplicate alignments:	no
Number of windows:	400
BAM file:	userspace/66ebdff874f1165c2d9d9eae8365cb5ac21cdeda/workflow/bsmap/F6703_372760_human.bam

2. Summary

2.1. Globals

Reference size	3,299,210,039
Number of reads	44,384,689
Mapped reads	44,384,689 / 100%
Unmapped reads	0 / 0%
Mapped paired reads	44,384,689 / 100%
Mapped reads, first in pair	22,205,447 / 50.03%
Mapped reads, second in pair	22,179,242 / 49.97%
Mapped reads, both in pair	44,356,998 / 99.94%
Mapped reads, singletons	27,691 / 0.06%
Secondary alignments	14,440,812
Read min/max/mean length	20 / 143 / 186.3
Duplicated reads (estimated)	35,998,499 / 81.11%
Duplication rate	51.22%
Clipped reads	3,049,431 / 6.87%

2.2. ACGT Content

Number/percentage of A's	1,468,908,389 / 23.65%
Number/percentage of C's	1,622,491,360 / 26.13%
Number/percentage of T's	1,473,114,146 / 23.72%
Number/percentage of G's	1,643,734,253 / 26.47%
Number/percentage of N's	1,627,225 / 0.03%
GC Percentage	52.6%

2.3. Coverage

Mean	22.1111
Standard Deviation	181.8145

2.4. Mapping Quality

Mean Mapping Quality	15.7

2.5. Insert size

Mean	5,718.87
Standard Deviation	37,381.38
P25/Median/P75	330 / 735 / 2,541

2.6. Mismatches and indels

General error rate	0.03%
Mismatches	20,453,013
Insertions	432,912
Mapped reads with at least one insertion	0.97%
Deletions	255,986
Mapped reads with at least one deletion	0.57%
Homopolymer indels	50.31%

2.7. Chromosome stats

Name	**Length**	**Mapped bases**	**Mean coverage**	**Standard deviation**

chr1	248956422	5924988767	23.7993	368.9212
chr10	133797422	2871374400	21.4606	145.329
chr11	135086622	3114111320	23.0527	154.2377
chr11_KI270721v1_random	100316	2600532	25.9234	32.4542
chr12	133275309	2733665693	20.5114	124.7041
chr13	114364328	1003448727	8.7741	75.9219
chr14	107043718	14266374409	133.2761	231.6792
chr14_GL000009v2_random	201709	0	0	0
chr14_GL000225v1_random	211173	231	0.0011	0.0331
chr14_KI270722v1_random	194050	0	0	0
chr14_GL000194v1_random	191469	1622984	8.4765	8.428
chr14_KI270723v1_random	38115	0	0	0
chr14_KI270724v1_random	39555	0	0	0
chr14_KI270725v1_random	172810	0	0	0
chr14_KI270726v1_random	43739	6273	0.1434	1.0172
chr15	101991189	1405638384	13.782	117.0957
chr15_KI2707	448248	852	0.0019	0.0436

27v1_random				
chr16	90338345	1996649446	22.1019	142.952
chr16_KI270728v1_random	1872759	3824	0.002	0.0376
chr17	83257441	2543226744	30.5465	159.2803
chr17_GL000205v2_random	185591	8965	0.0483	0.1285
chr17_KI270729v1_random	280839	0	0	0
chr17_KI270730v1_random	112551	0	0	0
chr18	80373285	512827480	6.3806	26.1123
chr19	58617616	3628657047	61.9039	253.6404
chr1_KI270706v1_random	175055	33503	0.1914	0.3324
chr1_KI270707v1_random	32032	0	0	0
chr1_KI270708v1_random	127682	0	0	0
chr1_KI270709v1_random	66860	0	0	0
chr1_KI270710v1_random	40176	7654	0.1905	0.2002
chr1_KI270711v1_random	42210	130595	3.0939	3.3398
chr1_KI270712v1_random	176043	0	0	0
chr1_KI27071	40745	1123	0.0276	0.2117

3v1_random				
chr1_KI270714v1_random	41717	0	0	0
chr2	242193529	7183080636	29.6584	166.3246
chr20	64444167	1713037812	26.5817	87.3048
chr21	46709983	617470404	13.2192	86.5261
chr22	50818468	1411511033	27.7756	79.0111
chr22_KI270731v1_random	150754	5313	0.0352	0.5984
chr22_KI270732v1_random	41543	0	0	0
chr22_KI270733v1_random	179772	15202411	84.565	578.3754
chr22_KI270734v1_random	165050	44500	0.2696	0.4533
chr22_KI270735v1_random	42811	0	0	0
chr22_KI270736v1_random	181920	0	0	0
chr22_KI270737v1_random	103838	0	0	0
chr22_KI270738v1_random	99375	0	0	0
chr22_KI270739v1_random	73985	0	0	0
chr2_KI270715v1_random	161471	0	0	0
chr2_KI270716v1_random	153799	0	0	0

chr3	198295559	3668569963	18.5005	170.0666
chr3_GL000221v1_random	155397	323007	2.0786	2.0729
chr4	190214555	1341956323	7.055	47.4048
chr4_GL000008v2_random	209709	283	0.0013	0.0367
chr5	181538259	2321409061	12.7874	53.7366
chr5_GL000208v1_random	92689	278	0.003	0.0547
chr6	170805979	1965852577	11.5093	175.3385
chr7	159345973	2270746864	14.2504	174.6186
chr8	145138636	2486768714	17.1337	276.6411
chr9	138394717	1632437347	11.7955	74.7582
chr9_KI270717v1_random	40062	0	0	0
chr9_KI270718v1_random	38054	4505	0.1184	0.4081
chr9_KI270719v1_random	176845	3370	0.0191	0.1592
chr9_KI270720v1_random	39050	0	0	0
chr1_KI270762v1_alt	354444	1851272	5.223	14.305
chr1_KI270766v1_alt	256271	0	0	0
chr1_KI270760v1_alt	109528	0	0	0
chr1_KI27076	185285	3576	0.0193	0.2207

5v1_alt				
chr1_GL383518v1_alt	182439	1139196	6.2443	9.5537
chr1_GL383519v1_alt	110268	859842	7.7977	20.0043
chr1_GL383520v2_alt	366580	20302409	55.3833	56.0957
chr1_KI270764v1_alt	50258	0	0	0
chr1_KI270763v1_alt	911658	1143623	1.2544	1.2984
chr1_KI270759v1_alt	425601	1125	0.0026	0.0745
chr1_KI270761v1_alt	165834	0	0	0
chr2_KI270770v1_alt	136240	0	0	0
chr2_KI270773v1_alt	70887	0	0	0
chr2_KI270774v1_alt	223625	0	0	0
chr2_KI270769v1_alt	120616	4941447	40.9684	40.98
chr2_GL383521v1_alt	143390	0	0	0
chr2_KI270772v1_alt	133041	0	0	0
chr2_KI270775v1_alt	138019	0	0	0
chr2_KI27077	110395	0	0	0

1v1_alt				
chr2_KI270768v1_alt	110099	0	0	0
chr2_GL582966v2_alt	96131	12441	0.1294	0.2011
chr2_GL383522v1_alt	123821	274982	2.2208	2.5893
chr2_KI270776v1_alt	174166	0	0	0
chr2_KI270767v1_alt	161578	760	0.0047	0.0905
chr3_JH636055v2_alt	173151	0	0	0
chr3_KI270783v1_alt	109187	0	0	0
chr3_KI270780v1_alt	224108	0	0	0
chr3_GL383526v1_alt	180671	0	0	0
chr3_KI270777v1_alt	173649	0	0	0
chr3_KI270778v1_alt	248252	0	0	0
chr3_KI270781v1_alt	113034	282	0.0025	0.0582
chr3_KI270779v1_alt	205312	0	0	0
chr3_KI270782v1_alt	162429	402	0.0025	0.0407
chr3_KI27078	184404	55556	0.3013	1.2333

4v1_alt				
chr4_KI270790v1_alt	220246	0	0	0
chr4_GL383528v1_alt	376187	0	0	0
chr4_KI270787v1_alt	111943	0	0	0
chr4_GL000257v2_alt	586476	30424	0.0519	0.1143
chr4_KI270788v1_alt	158965	0	0	0
chr4_GL383527v1_alt	164536	86709	0.527	1.4852
chr4_KI270785v1_alt	119912	0	0	0
chr4_KI270789v1_alt	205944	0	0	0
chr4_KI270786v1_alt	244096	560	0.0023	0.0677
chr5_KI270793v1_alt	126136	0	0	0
chr5_KI270792v1_alt	179043	275890	1.5409	3.0031
chr5_KI270791v1_alt	195710	29853	0.1525	0.883
chr5_GL383532v1_alt	82728	0	0	0
chr5_GL949742v1_alt	226852	0	0	0
chr5_KI27079	164558	0	0	0

4v1_alt				
chr5_GL339449v2_alt	1612928	7171908	4.4465	5.3234
chr5_GL383530v1_alt	101241	0	0	0
chr5_KI270796v1_alt	172708	0	0	0
chr5_GL383531v1_alt	173459	0	0	0
chr5_KI270795v1_alt	131892	7437	0.0564	0.4229
chr6_GL000250v2_alt	4672374	15428302	3.302	11.0233
chr6_KI270800v1_alt	175808	0	0	0
chr6_KI270799v1_alt	152148	0	0	0
chr6_GL383533v1_alt	124736	0	0	0
chr6_KI270801v1_alt	870480	1181305	1.3571	1.4332
chr6_KI270802v1_alt	75005	44919	0.5989	2.4229
chr6_KB021644v2_alt	185823	0	0	0
chr6_KI270797v1_alt	197536	233584	1.1825	1.6706
chr6_KI270798v1_alt	271782	1350295	4.9683	5.1486
chr7_KI27080	157952	28977	0.1835	1.0665

4v1_alt				
chr7_KI270809v1_alt	209586	170676	0.8143	2.0918
chr7_KI270806v1_alt	158166	2741	0.0173	0.299
chr7_GL383534v2_alt	119183	1677155	14.0721	17.736
chr7_KI270803v1_alt	1111570	16108058	14.4913	19.457
chr7_KI270808v1_alt	271455	0	0	0
chr7_KI270807v1_alt	126434	0	0	0
chr7_KI270805v1_alt	209988	0	0	0
chr8_KI270818v1_alt	145606	672	0.0046	0.0912
chr8_KI270812v1_alt	282736	282	0.001	0.0365
chr8_KI270811v1_alt	292436	0	0	0
chr8_KI270821v1_alt	985506	301508	0.3059	0.7274
chr8_KI270813v1_alt	300230	561	0.0019	0.0432
chr8_KI270822v1_alt	624492	232657	0.3726	0.3819
chr8_KI270814v1_alt	141812	0	0	0
chr8_KI27081	374415	564	0.0015	0.0549

0v1_alt				
chr8_KI270819v1_alt	133535	1550375	11.6103	14.4317
chr8_KI270820v1_alt	36640	0	0	0
chr8_KI270817v1_alt	158983	2570467	16.1682	46.3663
chr8_KI270816v1_alt	305841	3837018	12.5458	20.9713
chr8_KI270815v1_alt	132244	0	0	0
chr9_GL383539v1_alt	162988	709	0.0044	0.0726
chr9_GL383540v1_alt	71551	0	0	0
chr9_GL383541v1_alt	171286	0	0	0
chr9_GL383542v1_alt	60032	0	0	0
chr9_KI270823v1_alt	439082	0	0	0
chr10_GL383545v1_alt	179254	12509	0.0698	0.0874
chr10_KI270824v1_alt	181496	0	0	0
chr10_GL383546v1_alt	309802	44903372	144.9422	148.7841
chr10_KI270825v1_alt	188315	10020	0.0532	0.2093
chr11_KI2708	210133	1002656	4.7715	8.6605

32v1_alt				
chr11_KI270830v1_alt	177092	479	0.0027	0.0519
chr11_KI270831v1_alt	296895	4553972	15.3387	16.5098
chr11_KI270829v1_alt	204059	0	0	0
chr11_GL383547v1_alt	154407	0	0	0
chr11_JH159136v1_alt	200998	0	0	0
chr11_JH159137v1_alt	191409	0	0	0
chr11_KI270827v1_alt	67707	0	0	0
chr11_KI270826v1_alt	186169	0	0	0
chr12_GL877875v1_alt	167313	148456	0.8873	1.343
chr12_GL877876v1_alt	408271	29817	0.073	1.1392
chr12_KI270837v1_alt	40090	0	0	0
chr12_GL383549v1_alt	120804	0	0	0
chr12_KI270835v1_alt	238139	99319	0.4171	0.7491
chr12_GL383550v2_alt	169178	17382	0.1027	1.1313
chr12_GL383	138655	0	0	0

552v1_alt				
chr12_GL383553v2_alt	152874	0	0	0
chr12_KI270834v1_alt	119498	24691	0.2066	1.0062
chr12_GL383551v1_alt	184319	0	0	0
chr12_KI270833v1_alt	76061	0	0	0
chr12_KI270836v1_alt	56134	0	0	0
chr13_KI270840v1_alt	191684	0	0	0
chr13_KI270839v1_alt	180306	0	0	0
chr13_KI270843v1_alt	103832	0	0	0
chr13_KI270841v1_alt	169134	0	0	0
chr13_KI270838v1_alt	306913	1416	0.0046	0.0984
chr13_KI270842v1_alt	37287	568	0.0152	0.1293
chr14_KI270844v1_alt	322166	9714	0.0302	0.1408
chr14_KI270847v1_alt	1511111	15086877	9.984	91.1035
chr14_KI270845v1_alt	180703	0	0	0
chr14_KI2708	1351393	1667495711	1,233.9088	1,322.9075

46v1_alt				
chr15_KI270852v1_alt	478999	565	0.0012	0.0394
chr15_KI270851v1_alt	263054	1691	0.0064	0.1021
chr15_KI270848v1_alt	327382	38946	0.119	0.2888
chr15_GL383554v1_alt	296527	0	0	0
chr15_KI270849v1_alt	244917	334177288	1,364.4512	2,446.1102
chr15_GL383555v2_alt	388773	10120	0.026	0.0876
chr15_KI270850v1_alt	430880	1106035	2.5669	10.9909
chr16_KI270854v1_alt	134193	277404	2.0672	2.593
chr16_KI270856v1_alt	63982	0	0	0
chr16_KI270855v1_alt	232857	4654130	19.9871	33.1688
chr16_KI270853v1_alt	2659700	16071858	6.0427	8.4634
chr16_GL383556v1_alt	192462	1267011	6.5832	8.3939
chr16_GL383557v1_alt	89672	12192	0.136	0.5255
chr17_GL383563v3_alt	375691	852992	2.2705	5.7123
chr17_KI2708	391357	1935336	4.9452	5.2861

62v1_alt				
chr17_KI270861v1_alt	196688	9872563	50.194	58.5276
chr17_KI270857v1_alt	2877074	39883916	13.8627	25.1321
chr17_JH159146v1_alt	278131	1363824	4.9035	11.5469
chr17_JH159147v1_alt	70345	0	0	0
chr17_GL383564v2_alt	133151	0	0	0
chr17_GL000258v2_alt	1821992	4787883	2.6278	4.7002
chr17_GL383565v1_alt	223995	0	0	0
chr17_KI270858v1_alt	235827	0	0	0
chr17_KI270859v1_alt	108763	846	0.0078	0.1084
chr17_GL383566v1_alt	90219	7010	0.0777	0.4674
chr17_KI270860v1_alt	178921	1276826	7.1363	10.793
chr18_KI270864v1_alt	111737	0	0	0
chr18_GL383567v1_alt	289831	102701	0.3543	0.4925
chr18_GL383570v1_alt	164789	0	0	0
chr18_GL383	198278	841	0.0042	0.0869

571v1_alt				
chr18_GL383568v1_alt	104552	0	0	0
chr18_GL383569v1_alt	167950	0	0	0
chr18_GL383572v1_alt	159547	0	0	0
chr18_KI270863v1_alt	167999	73325	0.4365	4.1228
chr19_KI270868v1_alt	61734	259961116	4,210.9877	5,517.5923
chr19_KI270865v1_alt	52969	573354	10.8243	10.9337
chr19_GL383573v1_alt	385657	35076	0.091	0.1068
chr19_GL383575v2_alt	170222	87097	0.5117	0.5419
chr19_GL383576v1_alt	188024	281	0.0015	0.0386
chr19_GL383574v1_alt	155864	1120609	7.1897	7.3922
chr19_KI270866v1_alt	43156	19571	0.4535	0.4904
chr19_KI270867v1_alt	233762	191350	0.8186	1.2593
chr19_GL949746v1_alt	987716	7979115	8.0783	17.139
chr20_GL383577v2_alt	128386	0	0	0
chr20_KI2708	118774	3063	0.0258	0.2926

69v1_alt				
chr20_KI270871v1_alt	58661	0	0	0
chr20_KI270870v1_alt	183433	299172	1.631	2.463
chr21_GL383578v2_alt	63917	0	0	0
chr21_KI270874v1_alt	166743	0	0	0
chr21_KI270873v1_alt	143900	0	0	0
chr21_GL383579v2_alt	201197	0	0	0
chr21_GL383580v2_alt	74653	9422	0.1262	0.4553
chr21_GL383581v2_alt	116689	8086603	69.3005	72.5249
chr21_KI270872v1_alt	82692	33682	0.4073	3.2949
chr22_KI270875v1_alt	259914	19390	0.0746	1.2207
chr22_KI270878v1_alt	186262	1624	0.0087	0.1907
chr22_KI270879v1_alt	304135	7234610	23.7875	40.9393
chr22_KI270876v1_alt	263666	36215	0.1374	2.9413
chr22_KI270877v1_alt	101331	10186025	100.5223	121.9242
chr22_GL383	96924	913990	9.43	14.7949

583v2_alt				
chr22_GL383582v2_alt	162811	930142	5.713	6.7908
chrX_KI270880v1_alt	284869	361639	1.2695	1.5053
chrX_KI270881v1_alt	144206	0	0	0
chr19_KI270882v1_alt	248807	0	0	0
chr19_KI270883v1_alt	170399	0	0	0
chr19_KI270884v1_alt	157053	0	0	0
chr19_KI270885v1_alt	171027	0	0	0
chr19_KI270886v1_alt	204239	0	0	0
chr19_KI270887v1_alt	209512	9498	0.0453	0.4119
chr19_KI270888v1_alt	155532	0	0	0
chr19_KI270889v1_alt	170698	0	0	0
chr19_KI270890v1_alt	184499	0	0	0
chr19_KI270891v1_alt	170680	0	0	0
chr1_KI270892v1_alt	162212	0	0	0
chr2_KI27089	214158	6135	0.0286	0.245

4v1_alt				
chr2_KI270893v1_alt	161218	853	0.0053	0.0725
chr3_KI270895v1_alt	162896	20502	0.1259	0.1625
chr4_KI270896v1_alt	378547	1317044	3.4792	4.5154
chr5_KI270897v1_alt	1144418	2805812	2.4517	2.7226
chr5_KI270898v1_alt	130957	80955	0.6182	0.8325
chr6_GL000251v2_alt	4795265	320329821	66.8013	136.0181
chr7_KI270899v1_alt	190869	46063	0.2413	1.4059
chr8_KI270901v1_alt	136959	0	0	0
chr8_KI270900v1_alt	318687	0	0	0
chr11_KI270902v1_alt	106711	0	0	0
chr11_KI270903v1_alt	214625	0	0	0
chr12_KI270904v1_alt	572349	58512	0.1022	2.2528
chr15_KI270906v1_alt	196384	284	0.0014	0.0533
chr15_KI270905v1_alt	5161414	852242	0.1651	0.328
chr17_KI2709	137721	573	0.0042	0.072

07v1_alt				
chr17_KI270910v1_alt	157099	4525284	28.8053	31.1231
chr17_KI270909v1_alt	325800	24089042	73.9381	74.9865
chr17_JH159148v1_alt	88070	0	0	0
chr17_KI270908v1_alt	1423190	6201399	4.3574	6.4859
chr18_KI270912v1_alt	174061	0	0	0
chr18_KI270911v1_alt	157710	0	0	0
chr19_GL949747v2_alt	729520	6197152	8.4948	23.2827
chr22_KB663609v1_alt	74013	13182	0.1781	0.2111
chrX_KI270913v1_alt	274009	995500	3.6331	5.1299
chr19_KI270914v1_alt	205194	0	0	0
chr19_KI270915v1_alt	170665	0	0	0
chr19_KI270916v1_alt	184516	0	0	0
chr19_KI270917v1_alt	190932	1832	0.0096	0.0391
chr19_KI270918v1_alt	123111	0	0	0
chr19_KI2709	170701	0	0	0

19v1_alt				
chr19_KI270920v1_alt	198005	15532	0.0784	0.1277
chr19_KI270921v1_alt	282224	0	0	0
chr19_KI270922v1_alt	187935	0	0	0
chr19_KI270923v1_alt	189352	0	0	0
chr3_KI270924v1_alt	166540	0	0	0
chr4_KI270925v1_alt	555799	2266	0.0041	0.1251
chr6_GL000252v2_alt	4604811	162734068	35.34	245.5576
chr8_KI270926v1_alt	229282	0	0	0
chr11_KI270927v1_alt	218612	3044	0.0139	0.1251
chr19_GL949748v2_alt	1064304	379	0.0004	0.0162
chr22_KI270928v1_alt	176103	794560	4.5119	12.0863
chr19_KI270929v1_alt	186203	3763	0.0202	0.0577
chr19_KI270930v1_alt	200773	0	0	0
chr19_KI270931v1_alt	170148	0	0	0
chr19_KI2709	215732	0	0	0

32v1_alt				
chr19_KI270933v1_alt	170537	0	0	0
chr19_GL000209v2_alt	177381	0	0	0
chr3_KI270934v1_alt	163458	0	0	0
chr6_GL000253v2_alt	4677643	184487539	39.4403	155.0629
chr19_GL949749v2_alt	1091841	0	0	0
chr3_KI270935v1_alt	197351	9841	0.0499	0.067
chr6_GL000254v2_alt	4827813	309801223	64.1701	146.3532
chr19_GL949750v2_alt	1066390	379	0.0004	0.0162
chr3_KI270936v1_alt	164170	8419	0.0513	0.3162
chr6_GL000255v2_alt	4606388	323512489	70.2313	186.841
chr19_GL949751v2_alt	1002683	1804	0.0018	0.0167
chr3_KI270937v1_alt	165607	0	0	0
chr6_GL000256v2_alt	4929269	471675759	95.6888	236.8921
chr19_GL949752v1_alt	987100	75425282	76.411	224.562
chr6_KI27075	76752	40414	0.5266	6.0178

8v1_alt				
chr19_GL949753v2_alt	796479	11632322	14.6047	23.128
chr19_KI270938v1_alt	1066800	29047274	27.2284	66.2527
chrM	16569	300863213	18,158.1998	16,241.4885
chrUn_KI270302v1	2274	0	0	0
chrUn_KI270304v1	2165	0	0	0
chrUn_KI270303v1	1942	0	0	0
chrUn_KI270305v1	1472	0	0	0
chrUn_KI270322v1	21476	0	0	0
chrUn_KI270320v1	4416	0	0	0
chrUn_KI270310v1	1201	0	0	0
chrUn_KI270316v1	1444	0	0	0
chrUn_KI270315v1	2276	0	0	0
chrUn_KI270312v1	998	0	0	0
chrUn_KI270311v1	12399	0	0	0
chrUn_KI270317v1	37690	0	0	0

chrUn_KI270412v1	1179	0	0	0
chrUn_KI270411v1	2646	0	0	0
chrUn_KI270414v1	2489	0	0	0
chrUn_KI270419v1	1029	0	0	0
chrUn_KI270418v1	2145	0	0	0
chrUn_KI270420v1	2321	0	0	0
chrUn_KI270424v1	2140	0	0	0
chrUn_KI270417v1	2043	0	0	0
chrUn_KI270422v1	1445	0	0	0
chrUn_KI270423v1	981	0	0	0
chrUn_KI270425v1	1884	0	0	0
chrUn_KI270429v1	1361	0	0	0
chrUn_KI270442v1	392061	3727	0.0095	0.2845
chrUn_KI270466v1	1233	0	0	0
chrUn_KI270465v1	1774	0	0	0

chrUn_KI270467v1	3920	0	0	0
chrUn_KI270435v1	92983	0	0	0
chrUn_KI270438v1	112505	0	0	0
chrUn_KI270468v1	4055	0	0	0
chrUn_KI270510v1	2415	0	0	0
chrUn_KI270509v1	2318	0	0	0
chrUn_KI270518v1	2186	0	0	0
chrUn_KI270508v1	1951	0	0	0
chrUn_KI270516v1	1300	0	0	0
chrUn_KI270512v1	22689	0	0	0
chrUn_KI270519v1	138126	0	0	0
chrUn_KI270522v1	5674	0	0	0
chrUn_KI270511v1	8127	0	0	0
chrUn_KI270515v1	6361	0	0	0
chrUn_KI270507v1	5353	0	0	0

chrUn_KI270517v1	3253	0	0	0
chrUn_KI270529v1	1899	0	0	0
chrUn_KI270528v1	2983	0	0	0
chrUn_KI270530v1	2168	0	0	0
chrUn_KI270539v1	993	0	0	0
chrUn_KI270538v1	91309	0	0	0
chrUn_KI270544v1	1202	0	0	0
chrUn_KI270548v1	1599	0	0	0
chrUn_KI270583v1	1400	0	0	0
chrUn_KI270587v1	2969	0	0	0
chrUn_KI270580v1	1553	0	0	0
chrUn_KI270581v1	7046	0	0	0
chrUn_KI270579v1	31033	0	0	0
chrUn_KI270589v1	44474	0	0	0
chrUn_KI270590v1	4685	0	0	0

chrUn_KI270584v1	4513	0	0	0
chrUn_KI270582v1	6504	0	0	0
chrUn_KI270588v1	6158	0	0	0
chrUn_KI270593v1	3041	0	0	0
chrUn_KI270591v1	5796	0	0	0
chrUn_KI270330v1	1652	0	0	0
chrUn_KI270329v1	1040	0	0	0
chrUn_KI270334v1	1368	0	0	0
chrUn_KI270333v1	2699	0	0	0
chrUn_KI270335v1	1048	0	0	0
chrUn_KI270338v1	1428	0	0	0
chrUn_KI270340v1	1428	0	0	0
chrUn_KI270336v1	1026	0	0	0
chrUn_KI270337v1	1121	0	0	0
chrUn_KI270363v1	1803	0	0	0

chrUn_KI270364v1	2855	0	0	0
chrUn_KI270362v1	3530	0	0	0
chrUn_KI270366v1	8320	0	0	0
chrUn_KI270378v1	1048	0	0	0
chrUn_KI270379v1	1045	0	0	0
chrUn_KI270389v1	1298	0	0	0
chrUn_KI270390v1	2387	0	0	0
chrUn_KI270387v1	1537	0	0	0
chrUn_KI270395v1	1143	0	0	0
chrUn_KI270396v1	1880	0	0	0
chrUn_KI270388v1	1216	0	0	0
chrUn_KI270394v1	970	0	0	0
chrUn_KI270386v1	1788	0	0	0
chrUn_KI270391v1	1484	0	0	0
chrUn_KI270383v1	1750	0	0	0

chrUn_KI270393v1	1308	0	0	0
chrUn_KI270384v1	1658	0	0	0
chrUn_KI270392v1	971	0	0	0
chrUn_KI270381v1	1930	0	0	0
chrUn_KI270385v1	990	0	0	0
chrUn_KI270382v1	4215	0	0	0
chrUn_KI270376v1	1136	0	0	0
chrUn_KI270374v1	2656	0	0	0
chrUn_KI270372v1	1650	0	0	0
chrUn_KI270373v1	1451	0	0	0
chrUn_KI270375v1	2378	0	0	0
chrUn_KI270371v1	2805	0	0	0
chrUn_KI270448v1	7992	0	0	0
chrUn_KI270521v1	7642	0	0	0
chrUn_GL000195v1	182896	15008	0.0821	0.3491

chrUn_GL000219v1	179198	756217	4.22	4.5005
chrUn_GL000220v1	161802	33920315	209.6409	889.6511
chrUn_GL000224v1	179693	9146	0.0509	0.169
chrUn_KI270741v1	157432	45033	0.286	0.5402
chrUn_GL000226v1	15008	0	0	0
chrUn_GL000213v1	164239	0	0	0
chrUn_KI270743v1	210658	0	0	0
chrUn_KI270744v1	168472	10843	0.0644	0.3473
chrUn_KI270745v1	41891	47804	1.1412	2.1386
chrUn_KI270746v1	66486	0	0	0
chrUn_KI270747v1	198735	0	0	0
chrUn_KI270748v1	93321	283	0.003	0.055
chrUn_KI270749v1	158759	0	0	0
chrUn_KI270750v1	148850	282	0.0019	0.0435
chrUn_KI270751v1	150742	284	0.0019	0.0557

chrUn_KI270752v1	27745	0	0	0
chrUn_KI270753v1	62944	0	0	0
chrUn_KI270754v1	40191	0	0	0
chrUn_KI270755v1	36723	837	0.0228	0.2605
chrUn_KI270756v1	79590	0	0	0
chrUn_KI270757v1	71251	0	0	0
chrUn_GL000214v1	137718	0	0	0
chrUn_KI270742v1	186739	3267	0.0175	0.1791
chrUn_GL000216v2	176608	0	0	0
chrUn_GL000218v1	161147	4665714	28.9532	28.7321
chrX	156040895	1273671183	8.1624	95.8278
chrY	57227415	47289551	0.8263	7.0325
chrY_KI270740v1_random	37240	0	0	0
chr1_KQ031383v1_fix	467143	1144	0.0024	0.0733
chr1_KQ983255v1_alt	278659	568	0.002	0.0638
chr1_KN5383	305542	1232729	4.0346	52.0039

61v1_fix				
chr1_KQ458383v1_alt	349938	33246	0.095	0.7975
chr1_KN196473v1_fix	166200	123506	0.7431	8.8445
chr1_KZ208904v1_alt	166136	57389	0.3454	1.4643
chr1_KN196472v1_fix	186494	1306	0.007	0.1039
chr1_KZ208905v1_alt	140355	284	0.002	0.0449
chr1_KQ458382v1_alt	141019	0	0	0
chr1_KV880763v1_alt	551020	351174	0.6373	2.4142
chr1_KN196474v1_fix	122022	17210	0.141	1.1125
chr1_KN538360v1_fix	460100	0	0	0
chr1_KZ208906v1_fix	330031	11375	0.0345	0.3664
chr1_KQ458384v1_alt	212205	7025	0.0331	0.5073
chr2_KQ031384v1_fix	481245	2378127	4.9416	10.6063
chr2_KZ208907v1_alt	181658	0	0	0
chr2_KQ983256v1_alt	535088	3562	0.0067	0.1205
chr2_KZ2089	140361	42001	0.2992	0.4494

08v1_alt				
chr2_KN538363v1_fix	365499	772500	2.1135	23.1461
chr2_KN538362v1_fix	208149	283	0.0014	0.0368
chr3_KV766192v1_fix	411654	7464	0.0181	0.2634
chr3_KN196475v1_fix	451168	126102	0.2795	1.6744
chr3_KQ031385v1_fix	373699	13941	0.0373	0.4171
chr3_KN538364v1_fix	415308	3777797	9.0964	9.9016
chr3_KZ208909v1_alt	175849	0	0	0
chr3_KQ031386v1_fix	165718	0	0	0
chr3_KN196476v1_fix	305979	0	0	0
chr4_KQ090013v1_alt	90922	0	0	0
chr4_KQ090014v1_alt	163749	0	0	0
chr4_KQ090015v1_alt	236512	0	0	0
chr4_KV766193v1_alt	420675	0	0	0
chr4_KQ983257v1_fix	230434	0	0	0
chr4_KQ9832	205407	1223	0.006	0.097

58v1_alt				
chr5_KZ208910v1_alt	135987	4574	0.0336	0.1143
chr5_KN196477v1_alt	139087	0	0	0
chr5_KV575243v1_alt	362221	6352	0.0175	0.2029
chr5_KV575244v1_fix	673059	1703065	2.5303	10.7112
chr6_KZ208911v1_fix	242796	109524	0.4511	7.9475
chr6_KQ090017v1_alt	82315	0	0	0
chr6_KQ031387v1_fix	320750	0	0	0
chr6_KN196478v1_fix	268330	0	0	0
chr6_KQ090016v1_fix	245716	1347	0.0055	0.1169
chr6_KV766194v1_fix	139427	124248	0.8911	1.2288
chr7_KV880764v1_fix	142129	5660	0.0398	0.3004
chr7_KV880765v1_fix	468267	108307	0.2313	2.1257
chr7_KZ208912v1_fix	589656	436634	0.7405	0.8296
chr7_KZ208913v1_alt	680662	361791	0.5315	0.8873
chr7_KQ0313	179932	0	0	0

88v1_fix				
chr8_KZ208915v1_fix	6367528	111242809	17.4703	486.3133
chr8_KV880767v1_fix	265876	0	0	0
chr8_KV880766v1_fix	156998	0	0	0
chr8_KZ208914v1_fix	165120	1046847	6.3399	26.447
chr9_KQ090018v1_alt	163882	0	0	0
chr9_KQ090019v1_alt	134099	0	0	0
chr9_KN196479v1_fix	330164	30455	0.0922	0.3408
chr10_KN538367v1_fix	420164	0	0	0
chr10_KQ090020v1_alt	185507	11468	0.0618	0.247
chr10_KN196480v1_fix	277797	0	0	0
chr10_KQ090021v1_fix	264545	3035045	11.4727	12.1257
chr10_KN538366v1_fix	85284	848	0.0099	0.16
chr10_KN538365v1_fix	14347	0	0	0
chr11_KQ759759v1_fix	196940	7593	0.0386	0.4552
chr11_KN538	203552	0	0	0

368v1_alt				
chr11_KV766195v1_fix	140877	0	0	0
chr11_KQ090022v1_fix	181958	0	0	0
chr11_KN196481v1_fix	108875	289925	2.6629	8.2702
chr12_KQ090023v1_alt	109323	282	0.0026	0.0507
chr12_KZ208916v1_fix	1046838	192671	0.1841	0.1906
chr12_KN538369v1_fix	541038	361384	0.6679	8.3669
chr12_KN196482v1_fix	211377	0	0	0
chr12_KZ208918v1_alt	174808	0	0	0
chr12_KQ759760v1_fix	315610	851	0.0027	0.0737
chr12_KZ208917v1_fix	64689	3622973	56.006	62.4375
chr12_KN538370v1_fix	86533	0	0	0
chr13_KN538372v1_fix	356766	1522	0.0043	0.108
chr13_KQ090024v1_alt	168146	0	0	0
chr13_KN196483v1_fix	35455	0	0	0
chr13_KN538	148762	0	0	0

373v1_fix				
chr13_KQ090025v1_alt	123480	0	0	0
chr13_KN538371v1_fix	206320	20940	0.1015	0.404
chr14_KZ208920v1_fix	690932	524991	0.7598	1.3338
chr14_KZ208919v1_alt	171798	68575	0.3992	2.1737
chr15_KN538374v1_fix	4998962	1355064	0.2711	14.6443
chr15_KQ031389v1_alt	2365364	618189	0.2614	0.6403
chr16_KQ090026v1_alt	59016	2956	0.0501	0.3267
chr16_KV880768v1_fix	1927115	2984777	1.5488	9.8945
chr16_KQ090027v1_alt	267463	37080	0.1386	1.4905
chr16_KZ208921v1_alt	78609	1976	0.0251	0.2387
chr16_KQ031390v1_alt	169136	0	0	0
chr17_KV766196v1_fix	281919	8776561	31.1315	110.7681
chr17_KV575245v1_fix	154723	589622	3.8108	24.8832
chr17_KV766198v1_alt	276292	2816	0.0102	0.0738
chr17_KV766	246895	0	0	0

197v1_alt				
chr18_KQ458385v1_alt	205101	0	0	0
chr18_KQ090028v1_fix	407387	561	0.0014	0.0523
chr18_KZ208922v1_fix	93070	0	0	0
chr19_KQ458386v1_fix	405389	90074	0.2222	3.9869
chr19_KN196484v1_fix	370917	19563	0.0527	0.1103
chr19_KV575246v1_alt	163926	0	0	0
chr19_KV575247v1_alt	170206	0	0	0
chr19_KV575248v1_alt	168131	0	0	0
chr19_KV575249v1_alt	293522	527	0.0018	0.0423
chr19_KV575250v1_alt	241058	70417	0.2921	5.4289
chr19_KV575251v1_alt	159285	0	0	0
chr19_KV575252v1_alt	178197	0	0	0
chr19_KV575253v1_alt	166713	0	0	0
chr19_KV575254v1_alt	99845	242	0.0024	0.0492
chr19_KV575	161095	0	0	0

255v1_alt				
chr19_KV575256v1_alt	223118	0	0	0
chr19_KV575257v1_alt	100553	0	0	0
chr19_KV575259v1_alt	171263	0	0	0
chr19_KV575260v1_alt	145691	0	0	0
chr19_KV575258v1_alt	156965	850	0.0054	0.0678
chr22_KN196485v1_alt	156562	85108	0.5436	3.7618
chr22_KQ458387v1_alt	155930	25722	0.165	1.3168
chr22_KQ458388v1_alt	174749	1414	0.0081	0.1178
chr22_KN196486v1_alt	153027	3235	0.0211	0.1802
chr22_KQ759761v1_alt	145162	29593	0.2039	0.2296
chr22_KQ759762v1_fix	101037	5162	0.0511	0.3815
chrX_KV766199v1_alt	188004	0	0	0
chrY_KZ208923v1_fix	48370	0	0	0
chrY_KZ208924v1_fix	209722	0	0	0
chrY_KN1964	101150	0	0	0

87v1_fix				
chr1_KZ559100v1_fix	44955	0	0	0
chr3_KZ559104v1_fix	105527	0	0	0
chr3_KZ559105v1_alt	195063	21033	0.1078	0.7842
chr3_KZ559103v1_alt	302885	860315	2.8404	3.5055
chr3_KZ559102v1_alt	197752	0	0	0
chr3_KZ559101v1_alt	164041	0	0	0
chr7_KZ559106v1_alt	172555	0	0	0
chr8_KZ559107v1_alt	103072	0	0	0
chr11_KZ559109v1_fix	279644	194061	0.694	4.916
chr11_KZ559108v1_fix	305244	0	0	0
chr11_KZ559111v1_alt	181167	100873	0.5568	7.8312
chr11_KZ559110v1_alt	301637	0	0	0
chr12_KZ559112v1_alt	154139	517	0.0034	0.0675
chr16_KZ559113v1_fix	480415	127161	0.2647	0.3204
chr17_KZ559	116753	1134	0.0097	0.1364

114v1_alt				
chr18_KZ559116v1_alt	163186	0	0	0
chr18_KZ559115v1_fix	230843	0	0	0
chr2_ML143342v1_fix	84043	0	0	0
chr2_ML143341v1_fix	145975	0	0	0
chr3_ML143343v1_alt	215443	6996	0.0325	0.131
chr4_ML143344v1_fix	235734	0	0	0
chr4_ML143347v1_fix	176674	282	0.0016	0.0399
chr4_ML143346v1_fix	53476	0	0	0
chr4_ML143348v1_fix	125549	0	0	0
chr4_ML143345v1_fix	341066	0	0	0
chr4_ML143349v1_fix	276109	0	0	0
chr5_ML143350v1_fix	89956	2828	0.0314	0.2615
chr6_ML143351v1_fix	73265	0	0	0
chr7_ML143352v1_fix	254759	26162	0.1027	0.3973
chr9_ML1433	25408	0	0	0

53v1_fix				
chr10_ML143354v1_fix	454963	14771	0.0325	0.105
chr10_ML143355v1_fix	292944	818	0.0028	0.0654
chr11_ML143358v1_fix	270122	287785	1.0654	10.7349
chr11_ML143360v1_fix	170928	860662	5.0352	79.9866
chr11_ML143359v1_fix	217075	83053	0.3826	3.2004
chr11_ML143357v1_fix	165419	0	0	0
chr11_ML143356v1_fix	45257	0	0	0
chr12_ML143362v1_fix	192531	111574	0.5795	3.5003
chr12_ML143361v1_fix	297568	30316	0.1019	0.7594
chr13_ML143366v1_fix	409912	7636	0.0186	0.2609
chr13_ML143363v1_fix	7309	0	0	0
chr13_ML143364v1_fix	158944	283	0.0018	0.0514
chr13_ML143365v1_fix	65394	1127	0.0172	0.1826
chr14_ML143367v1_fix	399183	157383	0.3943	0.419
chr14_ML143	264228	283	0.0011	0.0327

368v1_alt				
chr15_ML143372v1_fix	396515	41358	0.1043	0.3799
chr15_ML143371v1_fix	5500449	201349	0.0366	0.9212
chr15_ML143370v1_fix	369264	39003	0.1056	0.1419
chr15_ML143369v1_fix	97763	84903	0.8685	3.0897
chr16_ML143373v1_fix	270967	6692	0.0247	0.3347
chr17_ML143374v1_fix	137908	7593	0.0551	0.1016
chr17_ML143375v1_fix	56695	283	0.005	0.0705
chr19_ML143376v1_fix	493165	450153	0.9128	5.4162
chr21_ML143377v1_fix	519485	1422139	2.7376	2.9458
chr22_ML143380v1_fix	412368	18135	0.044	0.3206
chr22_ML143378v1_fix	461303	0	0	0
chr22_ML143379v1_fix	12295	2256	0.1835	0.6846
chrX_ML143385v1_fix	17435	0	0	0
chrX_ML143382v1_fix	28824	0	0	0
chrX_ML1433	68192	0	0	0

83v1_fix				
chrX_ML143381v1_fix	403128	123979	0.3075	3.7004
chrX_ML143384v1_fix	14678	0	0	0
chr1_MU273333v1_fix	1572686	10459508	6.6507	10.6169
chr1_MU273330v1_alt	516764	28509	0.0552	0.5439
chr1_MU273335v1_fix	211934	367884	1.7358	7.7365
chr1_MU273336v1_fix	250447	0	0	0
chr1_MU273331v1_alt	847441	6725	0.0079	0.1131
chr1_MU273334v1_fix	210426	132705	0.6306	3.2262
chr1_MU273332v1_alt	335159	0	0	0
chr2_MU273344v1_fix	244725	0	0	0
chr2_MU273345v1_fix	174385	0	0	0
chr2_MU273343v1_fix	489404	13468	0.0275	0.0431
chr2_MU273340v1_alt	284971	502960	1.765	10.5176
chr2_MU273337v1_alt	431782	8444	0.0196	0.2699
chr2_MU2733	955087	136508	0.1429	1.5643

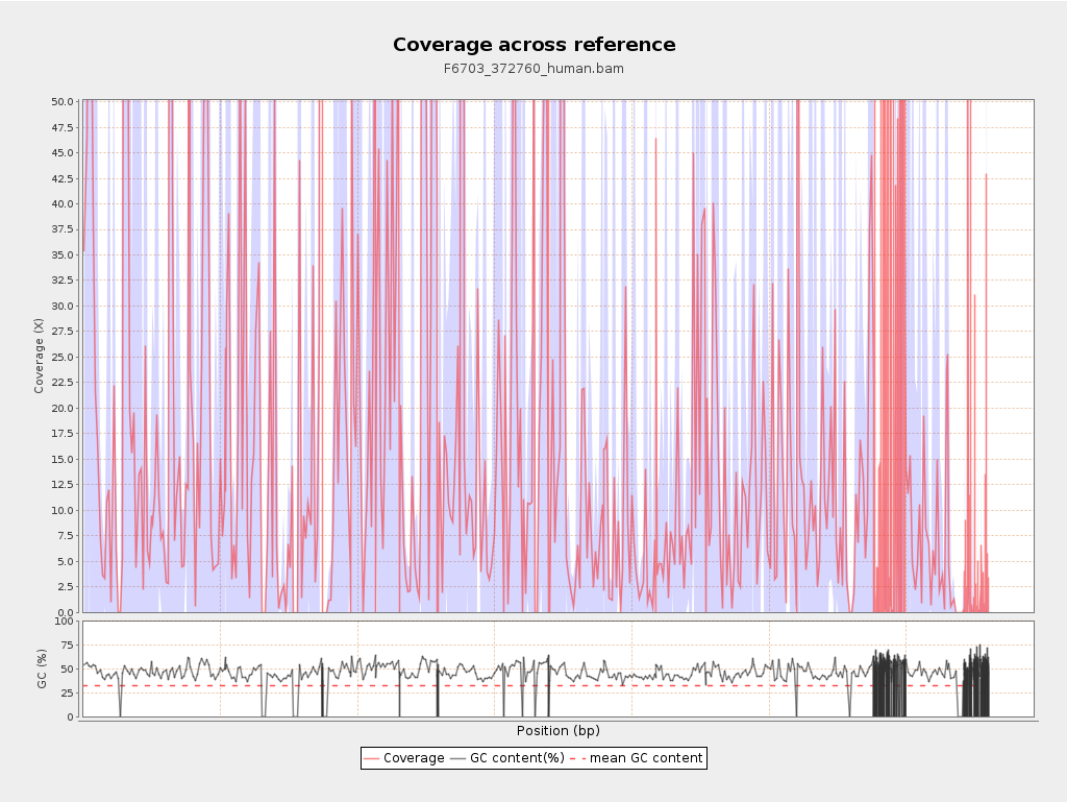
42v1_fix				
chr2_MU273339v1_alt	500581	188571	0.3767	5.6029
chr2_MU273338v1_alt	535251	2124281	3.9688	6.4533
chr2_MU273341v1_fix	120381	0	0	0
chr3_MU273347v1_fix	301310	104325	0.3462	0.3471
chr3_MU273348v1_fix	475876	284	0.0006	0.0252
chr3_MU273346v1_fix	469342	3681	0.0078	0.1928
chr4_MU273351v1_fix	205691	3076	0.015	0.3014
chr4_MU273349v1_alt	308682	34028	0.1102	2.0729
chr4_MU273350v1_fix	113364	0	0	0
chr5_MU273354v1_fix	2101585	329784	0.1569	2.6807
chr5_MU273353v1_fix	208405	56377	0.2705	3.5129
chr5_MU273356v1_alt	302485	0	0	0
chr5_MU273355v1_fix	508332	775	0.0015	0.0392
chr5_MU273352v1_fix	34400	0	0	0
chr6_MU2733	383128	353031	0.9214	7.7036

57v1_alt				
chr7_MU273358v1_alt	464417	0	0	0
chr8_MU273362v1_fix	429744	47364	0.1102	1.1691
chr8_MU273359v1_fix	150302	0	0	0
chr8_MU273361v1_fix	106905	332350	3.1088	18.3732
chr8_MU273363v1_fix	207371	30655	0.1478	0.2482
chr8_MU273360v1_fix	39290	5355	0.1363	0.6434
chr9_MU273366v1_fix	569668	23358	0.041	0.3375
chr9_MU273364v1_fix	340717	284	0.0008	0.0289
chr9_MU273365v1_fix	482250	20381	0.0423	0.3477
chr10_MU273367v1_fix	196262	0	0	0
chr11_KQ759759v2_fix	204999	4978	0.0243	0.2447
chr11_MU273369v1_fix	434831	5885231	13.5345	13.6185
chr11_MU273371v1_fix	122722	0	0	0
chr11_MU273370v1_fix	344606	264	0.0008	0.0391
chr11_MU273	261194	9689	0.0371	0.498

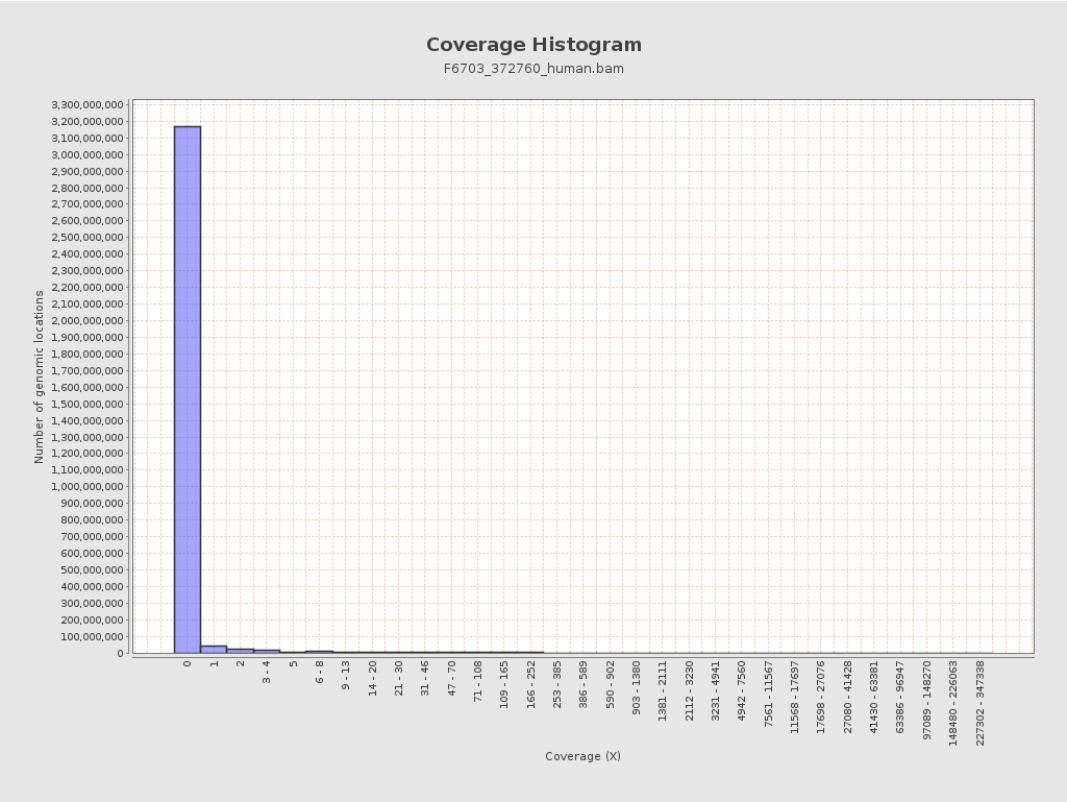
368v1_alt				
chr12_MU273372v1_fix	104537	790773	7.5645	13.4588
chr14_MU273373v1_fix	722645	6468	0.009	0.0714
chr15_MU273375v1_alt	204007	39212	0.1922	0.8784
chr15_MU273374v1_fix	1154574	43594	0.0378	0.1167
chr16_MU273376v1_fix	87715	3764884	42.9218	333.9763
chr16_MU273377v1_fix	334997	31983	0.0955	1.2286
chr17_MU273380v1_fix	538541	2480802	4.6065	56.8249
chr17_MU273378v1_alt	372839	2718	0.0073	0.0627
chr17_MU273383v1_fix	172609	993279	5.7545	11.5574
chr17_MU273379v1_fix	234878	1528	0.0065	0.072
chr17_MU273382v1_fix	187626	1071450	5.7106	64.2513
chr17_MU273381v1_fix	144689	1986	0.0137	0.2089
chr19_MU273386v1_fix	226166	232036	1.026	6.6532
chr19_MU273387v1_alt	89211	42442	0.4757	2.442
chr19_MU273	333754	6214	0.0186	0.2368

384v1_fix				
chr19_MU273385v1_fix	137818	0	0	0
chr20_MU273388v1_fix	273725	1127	0.0041	0.081
chr20_MU273389v1_fix	355731	133606	0.3756	3.2061
chr21_MU273390v1_fix	336752	69455	0.2062	1.5939
chr21_MU273391v1_fix	1020778	0	0	0
chr21_MU273392v1_fix	189707	364725	1.9226	5.5651
chr22_KQ759762v2_fix	101040	0	0	0
chrX_MU273397v1_alt	330493	1149814	3.4791	23.3847
chrX_MU273393v1_fix	68810	0	0	0
chrX_MU273394v1_fix	140567	34690	0.2468	1.7379
chrX_MU273396v1_alt	294119	0	0	0
chrX_MU273395v1_alt	619716	310234	0.5006	6.2947
chrY_MU273398v1_fix	865743	1128	0.0013	0.0853

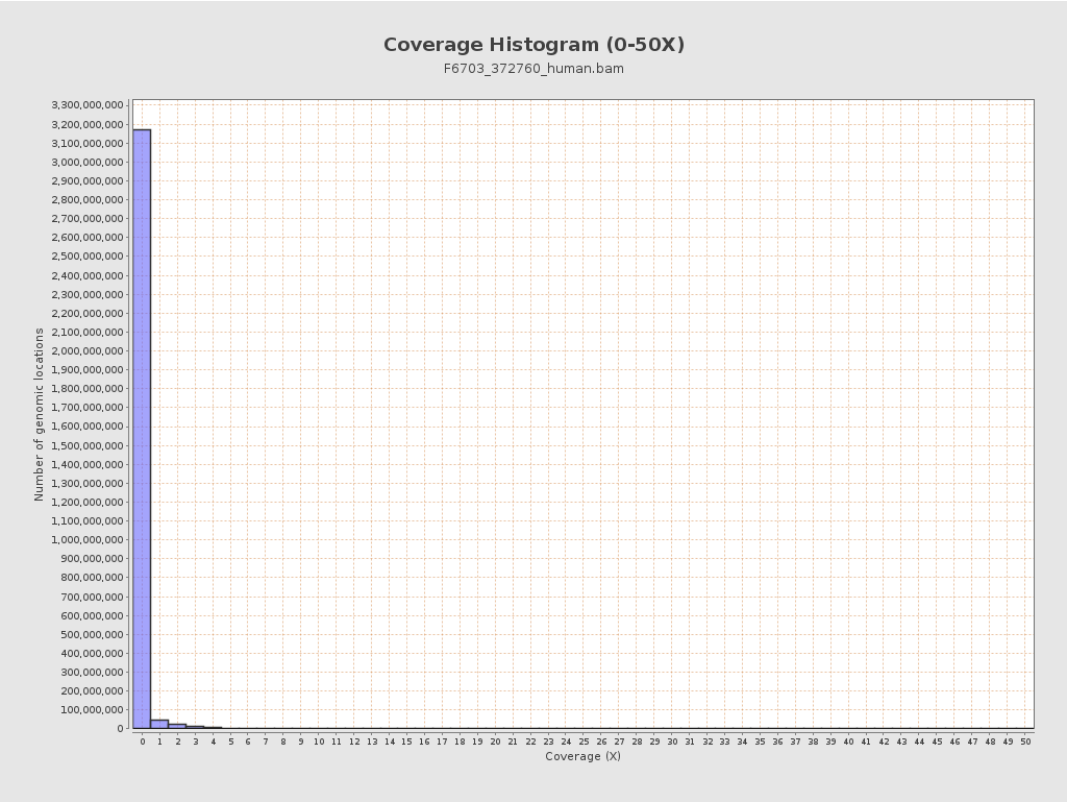
3. Results : Coverage across reference



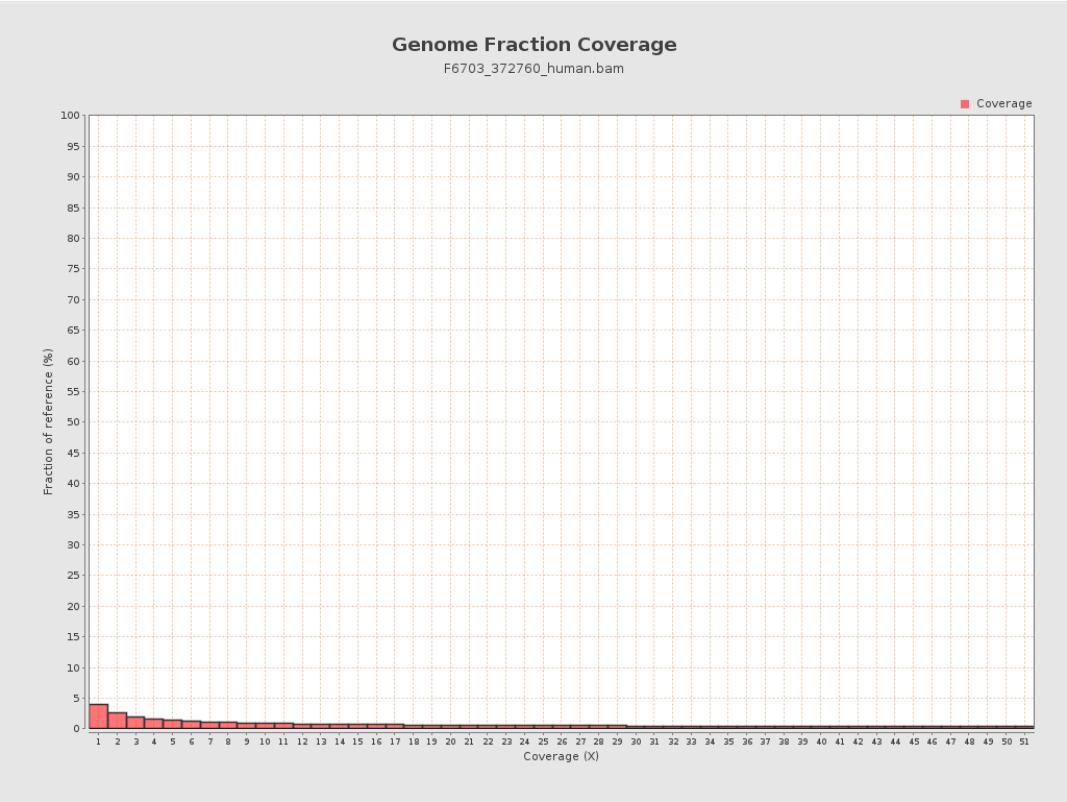
4. Results : Coverage Histogram



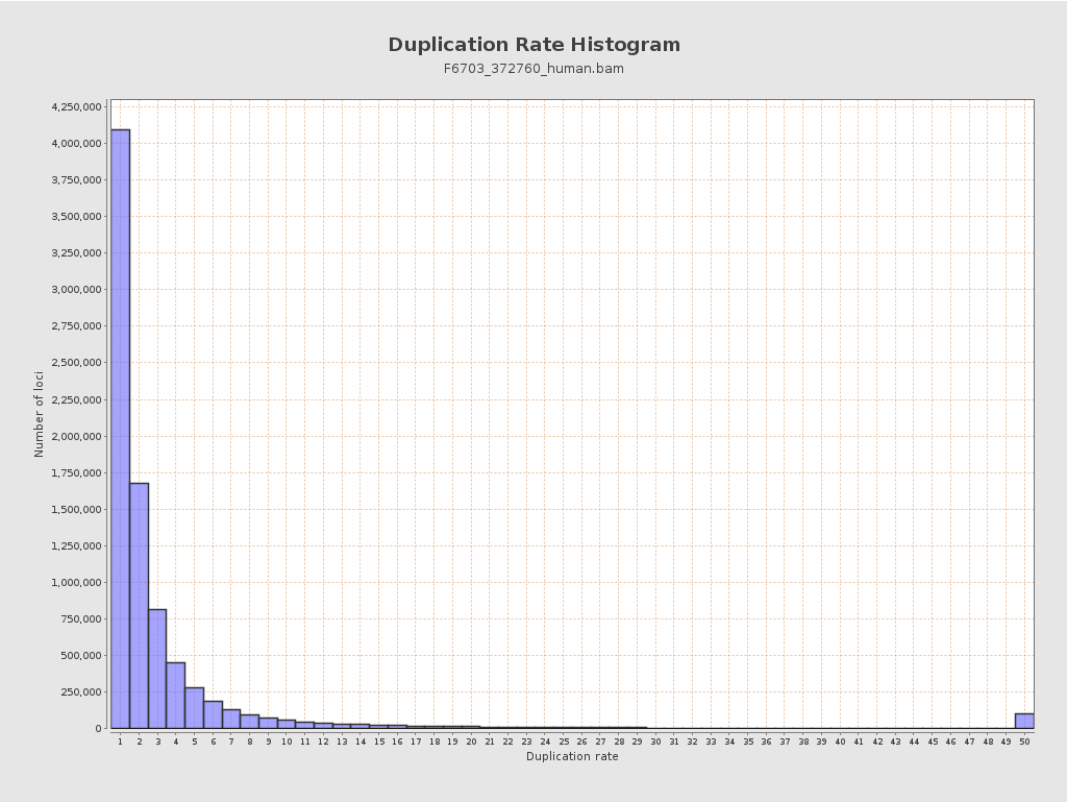
5. Results : Coverage Histogram (0-50X)



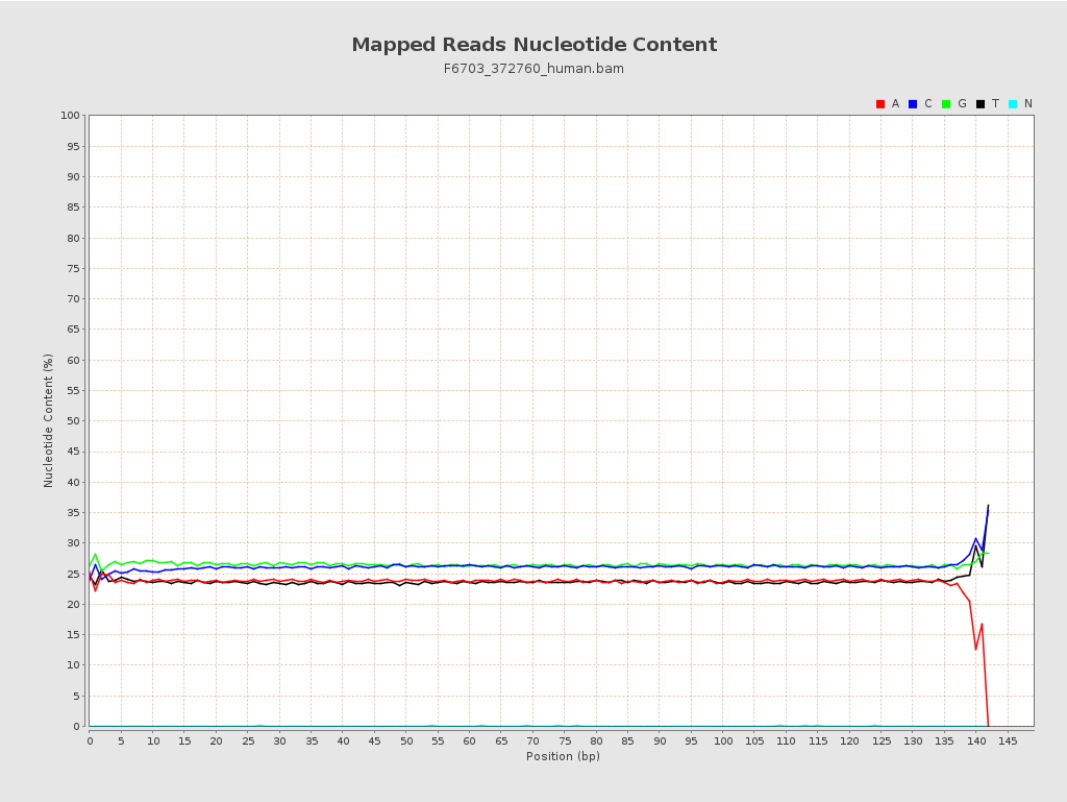
6. Results : Genome Fraction Coverage



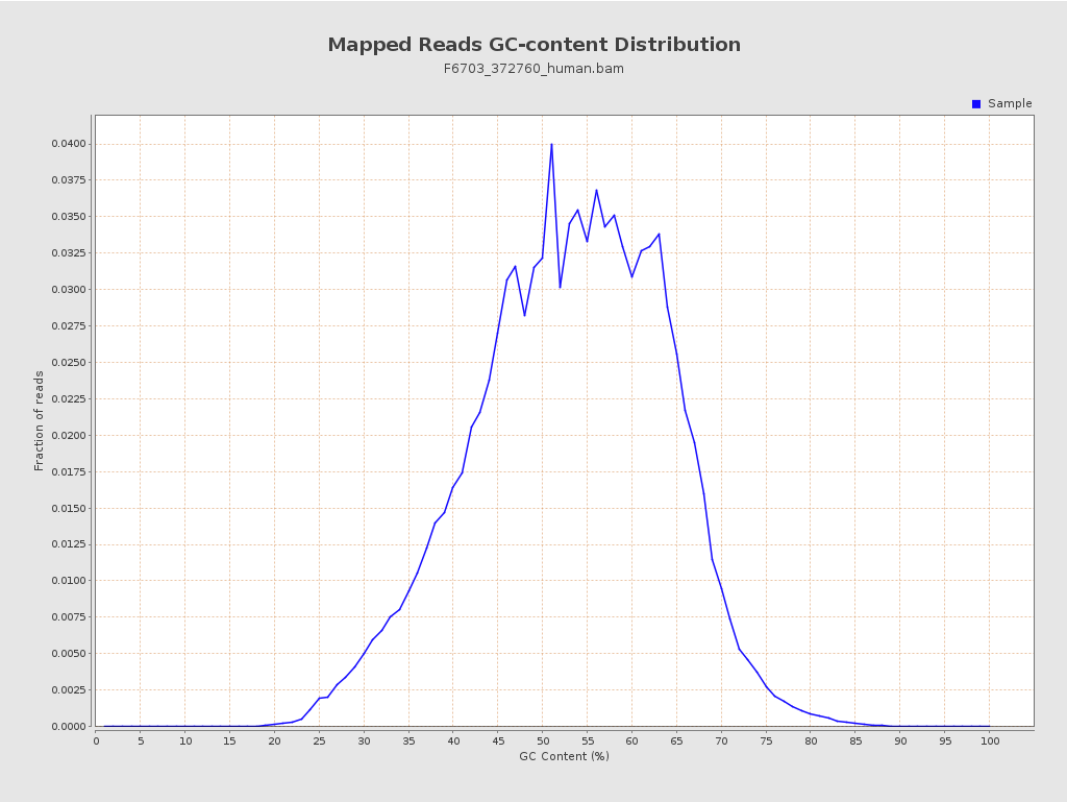
7. Results : Duplication Rate Histogram



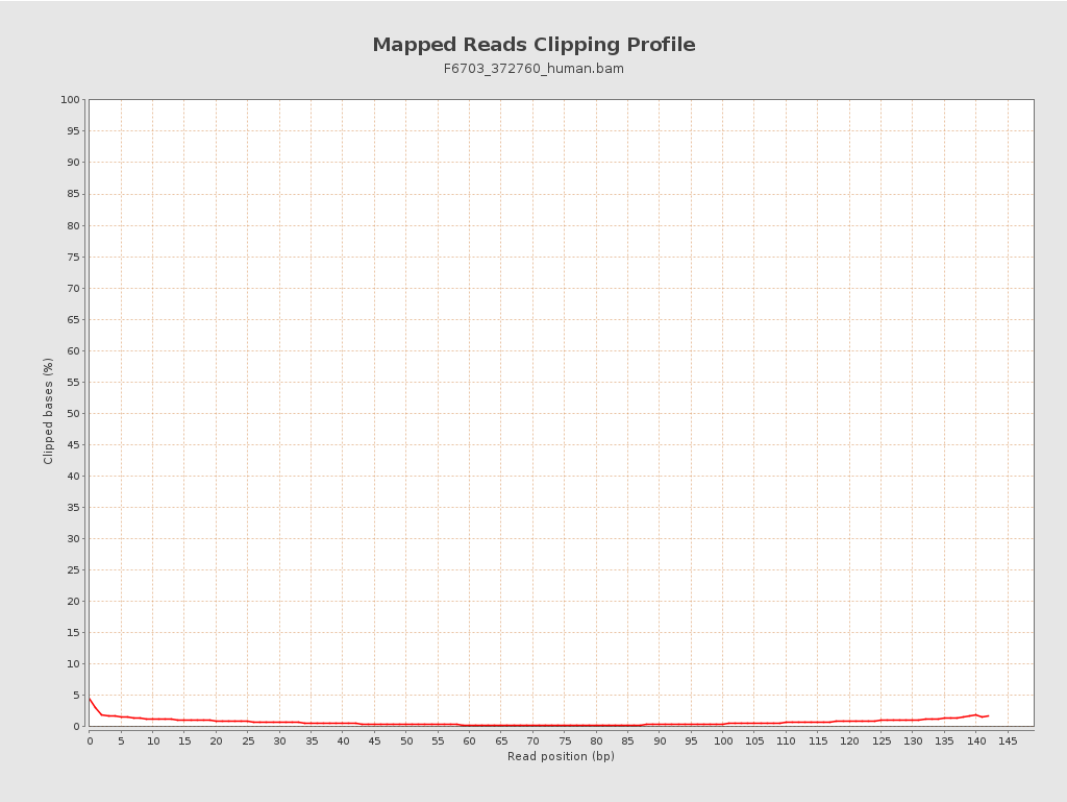
8. Results : Mapped Reads Nucleotide Content



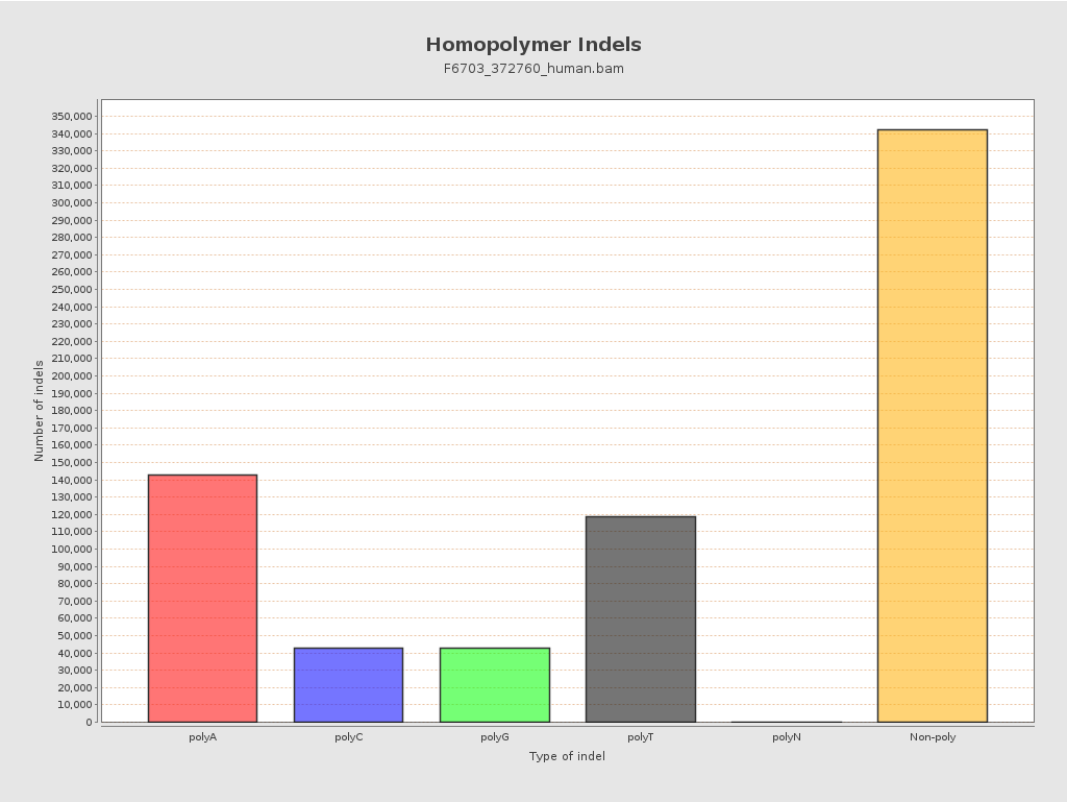
9. Results : Mapped Reads GC-content Distribution



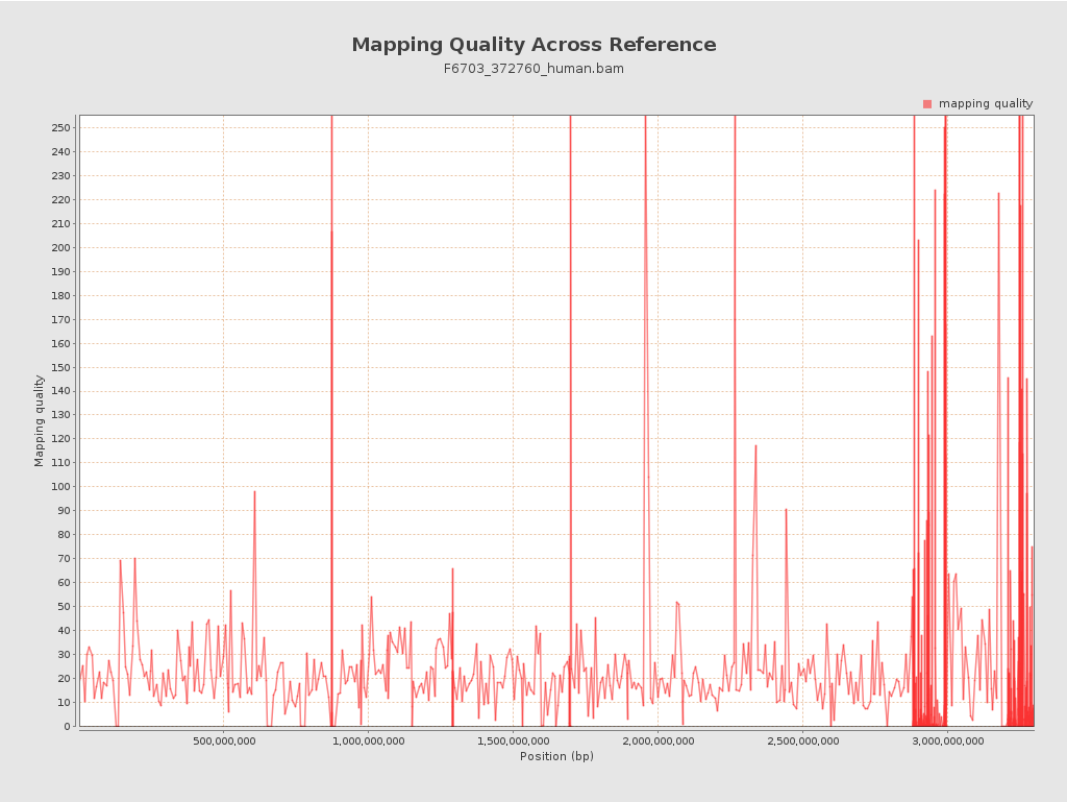
10. Results : Mapped Reads Clipping Profile



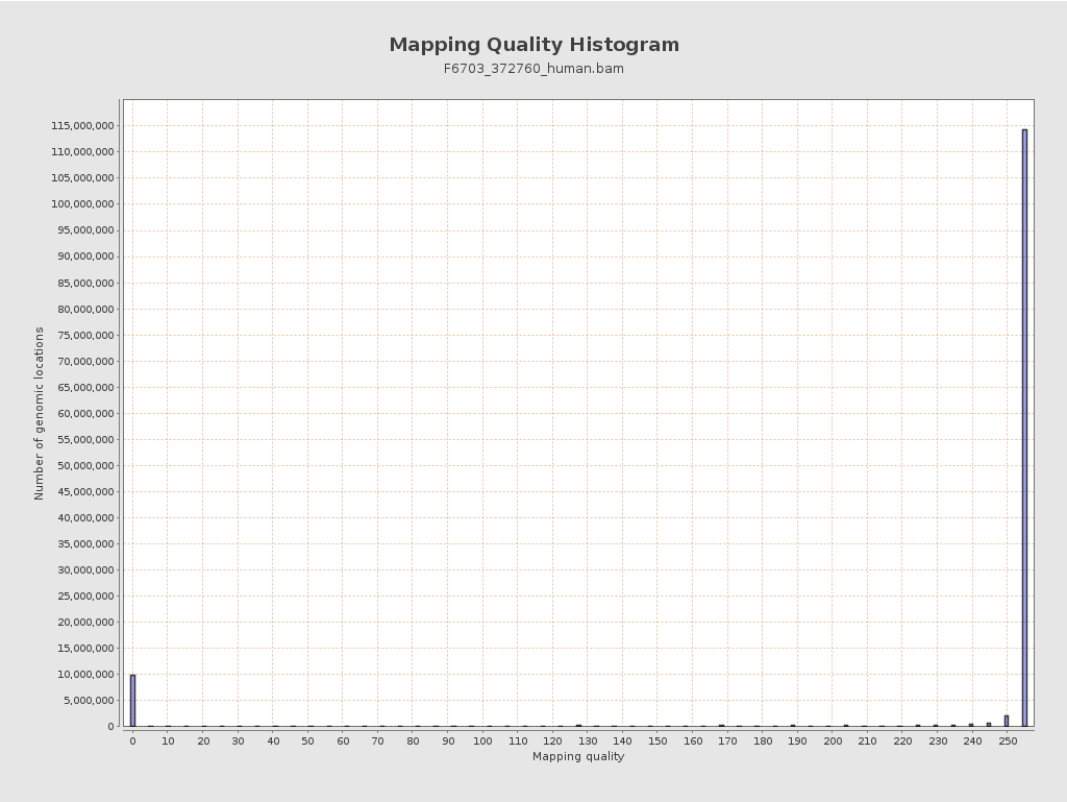
11. Results : Homopolymer Indels



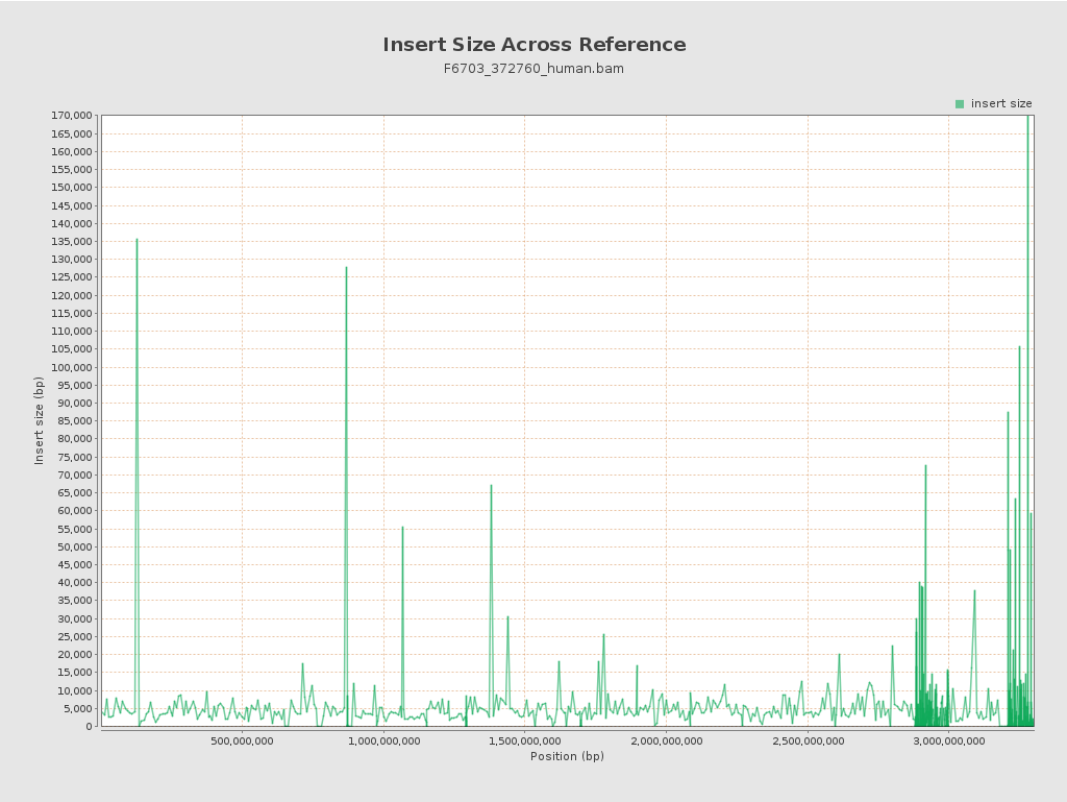
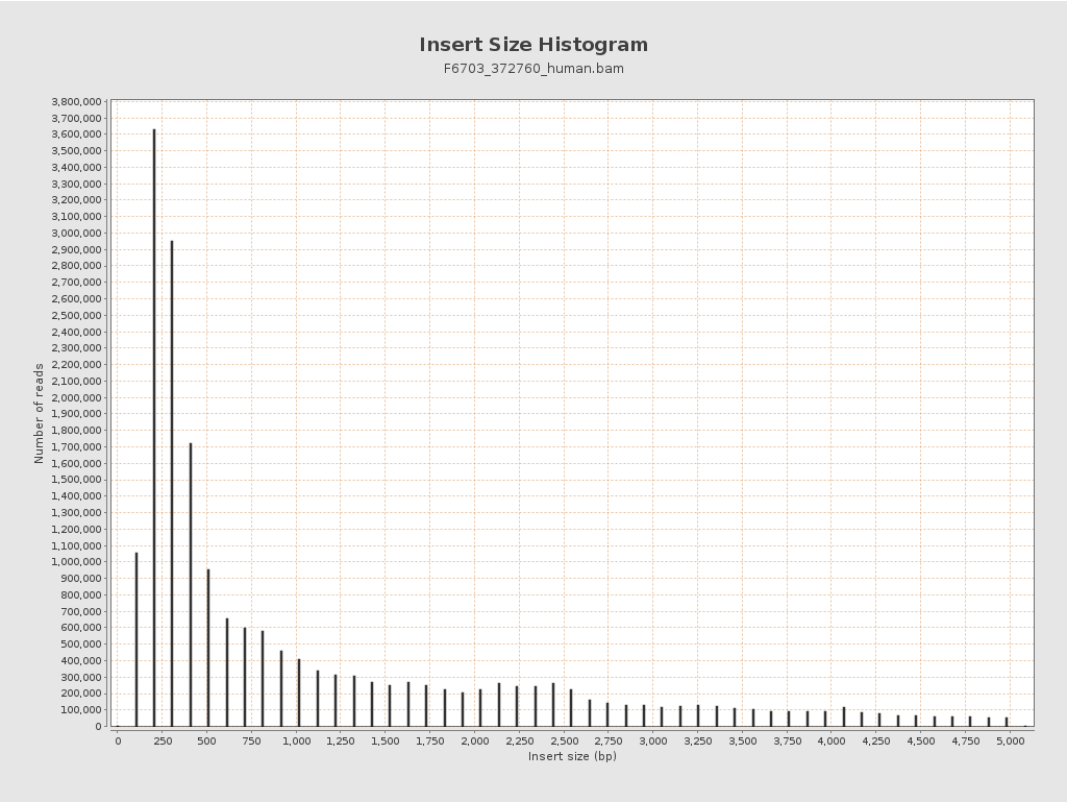
12. Results : Mapping Quality Across Reference



13. Results : Mapping Quality Histogram



14. Results : Insert Size Across Reference



15. Results : Insert Size Histogram